Introduction to Active Directory and Account Management – Part 2

Chapter 4

Groups

- The **best** ways to manage accounts is by grouping accounts that have similar characteristics
- When you use a group object as a security principal by adding it to the ACL of an object, all of the group's members receive the permissions that you assigned to the group.
- Groups also make it possible to assign permissions to multiple users at once.

Review of concepts

(check your understanding)

Inventory to review.

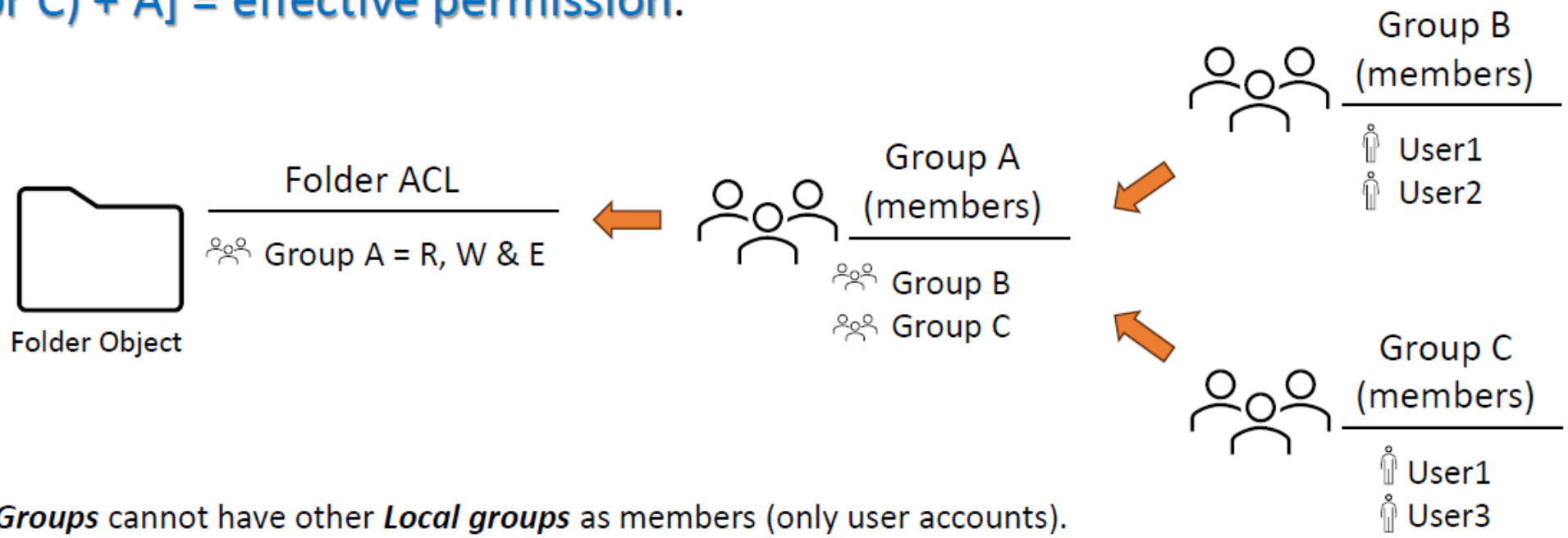
- What is a security principal?
- What is a User object?
- What is a Group object?
- What is an object (it reference to permissions)?
- What is an ACL?
- What is an ACE?
- What is an NTFS Permission?
- What is a permission?
- What is a policy?
- To which objects can a policy apply?
- What is a Computer/User object?
- What is a Right/Privilege?
- Local Security policy
- Group Policy

Group Scope

- Scope of influence (or scope)
  - Determines ***where*** in the network you can/should use the group
  - The *reach* of a group for gaining access to resources in Active Directory (access to resources in a single domain or access to all resources in all domains in a forest)
- Windows supports four group scopes:
  - Local ← This is the only type of group we used in MS I.
  - Domain local
  - Global
  - Universal

Group Nesting

- A group is a member of another group
  - Nesting reduces the number of times permissions must be assigned
- Example: If Group B, and Group C were nested in Group A then members of Group B and Group C would have the cumulative permissions of both groups [B (or C) + A] = effective permission.



Note: ***Local Groups*** cannot have other ***Local groups*** as members (only user accounts).

Types of Groups in AD

- Security groups
  - Define user rights & permissions to access resources and are listed in ACLs
  - Used to enable access to resources on a stand-alone server **or** in Active Directory
  - We will use Group Policy to assign user rights with security groups to delegate specific tasks
- Distribution groups
  - Used for e-mail or telephone lists, to provide quick, mass distribution of information
  - also known as a distribution list
  - There are no distribution groups if not joined to a domain.

Local ⟵
Domain local
Global
Universal

Local Groups

- Created using the Local Users and Computers (lusrmgr.msc) MMC Snap-In
- Can contain local user accounts, domain user and domain global groups from the domain or any trusted domain.
- Cannot have other local groups as members
- Can only be assigned permissions to resources on the local machine
- **Can not be created on DCs**

~~Local~~

Domain local

Global

Universal

Implementing Domain Local Groups

~~Local~~
Domain local
Global
Universal

- Domain local security group
  - Used when Active Directory is deployed
  - Typically used to **manage resources** in a domain and to give global groups from the same and other domains access to those resources
- The scope of a domain local group is the domain in which the group exists
- The typical purpose of a domain local group is to provide access to resources
  - You grant access to servers, folders, shared folders, and printers to a domain local group
- Best Practice: put domain local groups in ACLs **only**
  - Add global groups as members of the domain local.

Domain Local Groups

- Can add usersfrom any domain in the forest
- Can add global groups from any domain in the forest
- Can add domain local groups from the same domain only (which makes sense!)
- Can assign permissions to resources located only within the domain the group was created in (because this is the point of the group)
- Can be ***converted*** in Universal Group scope if the group doesn't coin any other Domain Local group.
- Created using Active Directory Users and Computers (not available outside of a domain environment)

~~Local~~

~~Domain local~~

Global ⟵

Universal

Implementing Global Groups

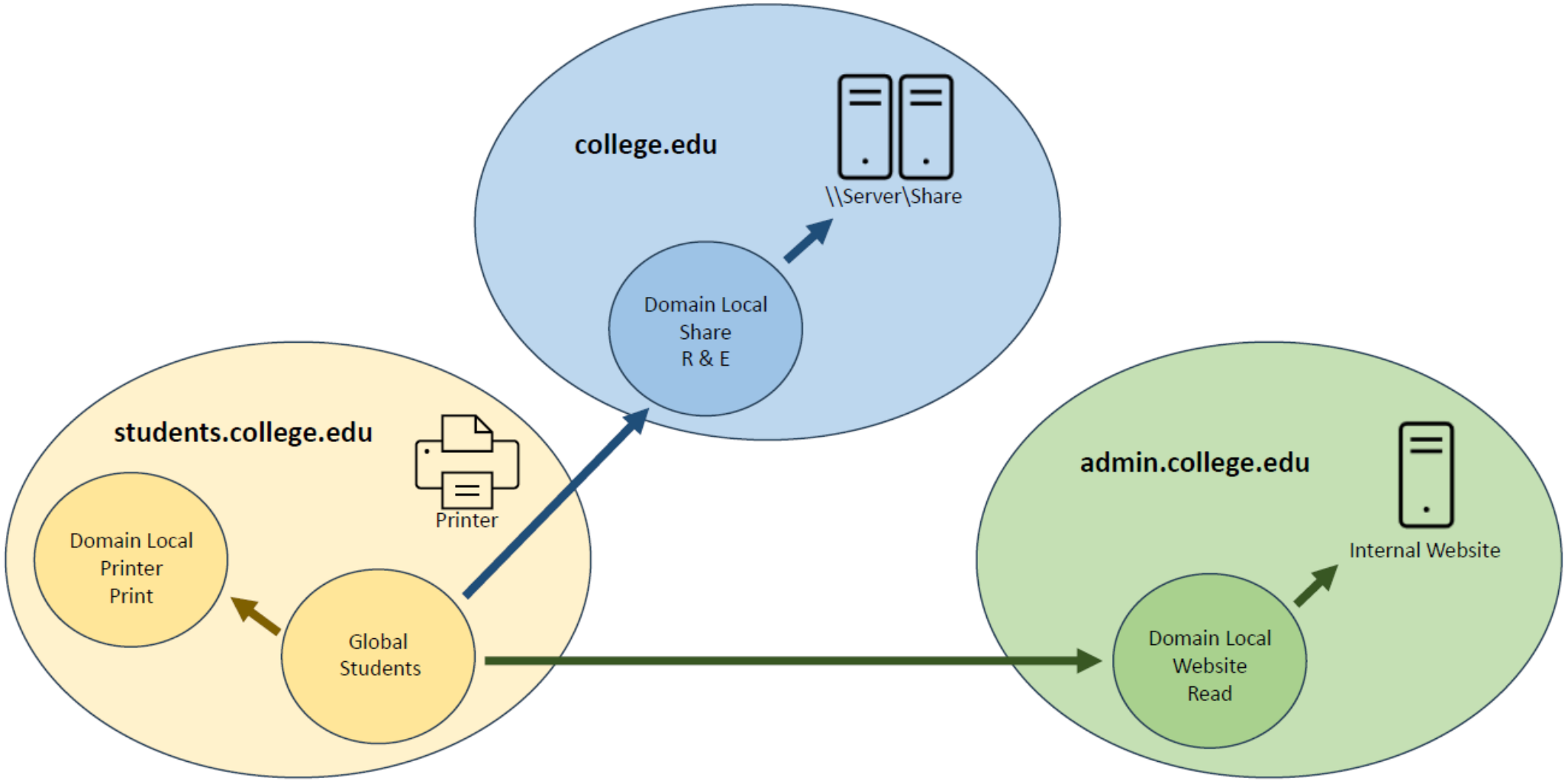
~~Local~~
~~Domain local~~
Global ⟸
Universal

- Global security group
  - Intended to contain user accounts from a single domain
  - Can also be set up as a member of a domain local group in the same or another domain
- A global group can contain user accounts and other global groups from the domain in which it was created
- A global group can be ***converted*** to a universal group
  - (As long as it is not nested in another global group or in a universal group)

Global Groups

- Can add users *only* from the same domain where the group was created
- Can add global groups from the same domain where the group was created
- Can be used to assign permissions to resources in any domain in the forest and any trusted domains.
- Created using Active Directory Users and Computers
- FYI: Global group membership is not replicated outside of its domain. This minimizes the amount of replication traffic

Implementing Global Groups



~~Local~~

~~Domain local~~

~~Global~~

Universal

Implementing Universal Groups

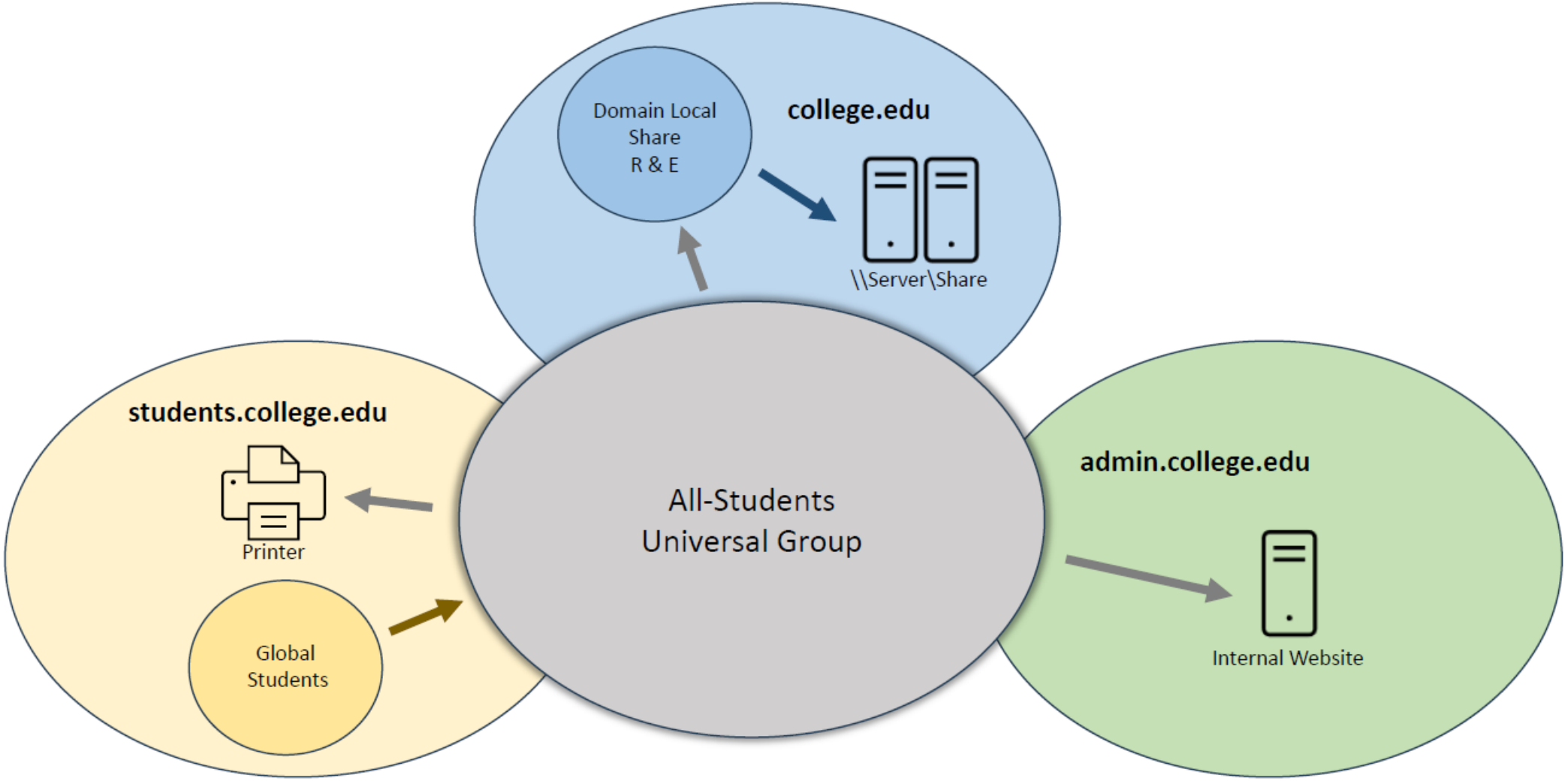
~~Local~~
~~Domain local~~
~~Global~~
Universal

- Universal security groups
  - Provide a means to span domains and trees
- Universal groups are offered to provide an easy means to access any resource in a tree

Universal Groups

- Can add users from any domain in the forest
- Can add global groups from any domain in the forest ⟸ GGs can best nested anywhere
- Can add other universal groups as members
- *Can* be a member of Domain Local Groups ⟸ DLGs can have members from anywhere
- *Cannot* have Domain local Groups **as Members.** ⟸ Breaks DLGs scope rule
- *Cannot* **be a Member** of Global Groups ⟸ Breaks GGs membership rule
- Can give permissions to domain resources in any domain in the forest or in trusted forests. (*only done where it makes sense, but not typically*)
- Used to provide easy access to objects in the tree
- Created using Active Directory Users and Computers
- Universal group membership ***is*** replicated to the Global Catalog.
  - Can easy the burden of management ***at the cost*** of increased replication traffic.

Implementing Universal Groups



~~Local~~

~~Domain local~~

~~Global~~

~~Universal~~

When should I use X group?

- Guidelines to help simplify how you plan to use groups:
  - Use global groups to hold accounts as members (never use in ACLs)
  - Use domain local groups to provide access to resources in a specific domain (only use on ACLs and have other groups as members)
  - Use universal groups to provide extensive access to resources (rarely used in practice, but sometimes req's)
- A global group is used to provide members with access to resources in the same or in another domain
  - ...then make the global group in one domain a member of a domain local group in the same or another domain
- The Domain Local group is used to provide members access to *this* domain from within *that* domain or from other domains in the forest

Converting Group Scopes

- Universal group can be changed to domain local, as long as it's not a member of a universal group
- Universal group can be changed to global, as long as it does not have a universal group as its member
- Global group can be changed to a universal group, provided its not a member of another global group
- Domain local group can be changed to a universal group, provided no domain local group is a member

Microsoft's Recommended Strategy for Groups in a Multidomain Environment

- Avoid the use of universal groups
- Create global groups to represent a grouping of users who will have consistent access requirements for a variety of resources (typically based on job titles or activities performed). Add the appropriate users to the global group.
  - When these users span multiple domains, multiple global groups will need to be created, since one global group can not have memberships that span multiple domains.
- Identify resources that users need to access. Create domain local groups based on the resource and the level of access the users need (example: users need print access to the Printer1, so I create a domain local group called Printer1-Print).
- Add the global groups that require permission to the domain local group that will be given permission.
- Assign the appropriate permission levels to the domain local groups on the resources they need to access.

Group Scope Summary

Scope	Possible members	Scope conversion	Can grant permissions	Possible member of
Local	Local Users, All Domain Groups	N/A	Only on the local computer	N/a
Domain Local	Universal (forest), Global (any domain), Domain Local (same domain only)	Universal scope if group doesn't contain a Domain Local group	Within the same domain	Domain Local (same domain only), Local groups on computers in the same domain (minus well-known).
Global	Accounts (same domain), Global (same domain)	Can be converted to Universal scope if self is not a member of Global/Universal group	Any domain in the same forest	Universal (same forest), Global (same domain), Domain Local (same forest)
Universal	Accounts (same forest), Global groups (same forest), Universal groups (same forest)	Domain Local scope (if group isn't a member of any a Universal), Global (if group doesn't contain a Universal group).	On any domain within the same forest	Universal (same forest), Domain Local groups (same forest), Local groups (same forest)

Default Built-In Local Groups

- Present on Windows Server stand-alone and member servers
- Membership gives a user the right to perform system tasks on a single computer, such as backing up & restoring files, or changing the system time
- Managed using the Local Users and Groups MMC snap-in

Group	Purpose
Administrators	•Complete and unrestricted access •Built-in Administrator account is a member •When clients join domain, Windows adds Domain Admins to the built-in Administrators group
Backup Operators	•Have the ability to override security restrictions for the purposes of backing up and restoring files

Predefined Active Directory Groups

- Security groups, (mostly global), intended to group common types of domain user accounts
- By default, these groups do not have any inherent rights or permissions

Group	Purpose
Domain Admins	•Member of the local Administrators group on domain member systems
Domain Computers	All computers in the domain, except DCs
Enterprise Admins	•Only created in the forest root domain •Only one in the forest. •Becomes a member of the Administrators domain local group in every domain in the forest
Group Policy Creator Owners	•Permitted to modify group policy settings in the domain •By default, domain Administrator is a member

Built-In Active Directory Groups

- Domain local security groups which are in the Built-in container in Active Directory
- Provide rights/permissions for members to perform tasks on DCs and in the AD tree

Needed more space
Next slide…

Built-In Active Directory Groups (cont's)

Group	Purpose
Account Operators	•Can create/delete/modify users, groups and computer objects in the Users and Computers containers and in all OUs, except Domain Controllers •Can not modify Administrators or Domain Admins groups •Can not modify accounts in the Administrators or Domain Admins groups •Can logon to domain controllers and shut down
Administrators	•Full administrative access to all DCs and to the domain itself •By default, Domain Admins and Enterprise Admins groups and the domain Administrator account are members of this group
Remote Desktop Users	Can remotely logon to domain controllers using Remote Desktop Services
Server Operators	Can logon, create and delete shared resources, stop/start services, backup and restore files, format the hard disk, and shutdown the server
Event Log Readers	Members are permitted to read event logs from all DCs
Performance Log Users	Can schedule logging of performance counters
Performance Monitor	Members have remote access to monitor using Performance Monitor

Special Identities

- Exist on all computers running Windows
- Not *technically* groups (cannot delete them, cannot create them, and cannot directly modify their memberships)
- Used in the same way groups are

Special Identity	Function
Anonymous Logon	Anyone who connects *without* authenticating
AuthenticatedUsers	All authenticated users with valid user account, *except* Guest
CreatorOwner	User who created or took ownership of a resource
Everyone	Authenticated Users plus the Guest account (includes anonymous on pre-2008)
Interactive	Users who are currently logged on locally or through Remote Desktop Services
Network	Users who have logged on using a network connection (not Remote Desktop)

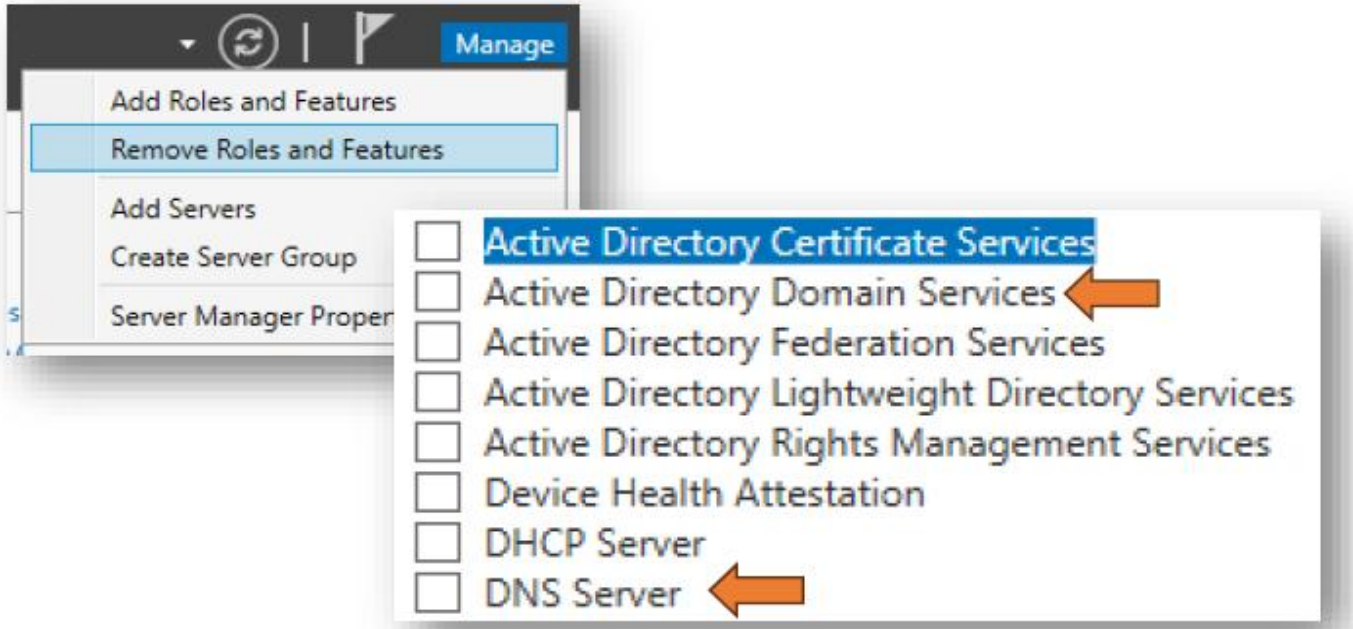
Creating Groups

- To create a group object, right-click a container\OU in ADUC → New, → Group
- Group Properties Tabs:
  - General
    - Used to enter a description, set the scope, and set the group type
  - Members
    - Used to add group members
  - Member Of
    - Used to join another group
  - Managed By
    - Establishes who will manage the group

Deleting Groups

- As with user objects, each group object that you create in AD has a unique, non-reusable SID. Windows Server uses the SID to identify the group and the permissions assigned to it.
- When you delete a group, Windows does not use the same group SID again, even if you create a group with the same name as the one you deleted.
- You cannot restore the access permissions you assigned to resources by recreating a deleted group object. You must add the newly created group as a security principal in the resource's ACL all over again.

Do this…



On Server2

- Uninstall AD DS
- Uninstall DNS
- Reboot.
- Set Server2's preferred DNS to the IP address of Server1

Naming Groups

Managing Permissions

AGUDLP

Letter	Layer	Role / Purpose
A	**Accounts**	The individual user or computer objects.
G	**Global Groups**	**Role-based.** Groups users from a *single domain* into a job role (e.g., `G_Sales_Users`).
U	**Universal Groups**	**The Bridge.** Consolidates Global groups from *multiple domains* across the entire forest (e.g., `U_Global_Sales`).
DL	**Domain Local**	**Resource-based.** Created in the domain where the resource sits to define an access level (e.g., `DL_FileServer_Read`).
P	**Permissions**	The actual NTFS or Share permissions applied to the **Domain Local** group.

AGUDLP Example

If you apply these naming rules your "permissions chain" looks like this:

- **A**ccount: `jdoe`
- **G**lobal Group: `G_Sales`
- **U**niversal Group: `U_Global_Sales`
- **DL** Domain Local: `DL_SalesFolder_Modify`
- **P**ermissions: *(Applied to the folder)*

AGUDLP Benefits

- **Reduced Replication:** Adding/removing users from a Global group doesn't trigger forest-wide replication (unlike Universal groups).
- **Clear Auditing:** To see who has access to a folder, you only look at one ***Domain Local*** group.
- **Token Optimization:** It prevents "token bloat" (where a user's login token becomes too large because they are in too many individual groups).

Prefix Based Naming

- Modern administration best-practices use a **standardized prefix** to indicate the group's "***Scope***" and "***Type***.“
  - This allows admins to filter thousands of groups in seconds using PowerShell or the GUI
- Microsoft does not enforce one single "mandatory" naming convention, but they have officially promoted the **AGUDLP** model for a long time
  - As well as a set of industry-standard prefixes that most senior admins use in Server 2022.
- The goal of a naming convention is to tell an admin three things just by looking at the name: Scope, Department/Role, and Access Level.

Prefix Based Naming - Scope

Scope Prefixes (Based on AGUDLP)

- **G_ (Global):** Used for Job Roles (Who).
  - *Example:* `G_HR_Recruiters`
- **U_ (Universal):** Used for Enterprise-wide roles (Cross-Domain).
  - Example: `U_All_Employees`
- **DL_ (Domain Local):** Used for Resource Access (What).
  - Example: `DL_FileServer_Accounting_Read`

Prefix Based Naming - Type

Type Prefixes (Based on AGUDLP)

- **`SEC_` (Security):** For permissions.
- **`DIST_` (Distribution):** For email lists only.
- **`ADM_` (Administrative):** For highly privileged groups (e.g., Domain Admins).

Prefix Based Naming - Examples

Most modern Windows Server environments use a combination of **Scope** (where it lives) and **Type** (what it does).

Prefix	Scope	Type	Use Case
`G_SEC_`	Global	Security	**The "Who":** Grouping users by job role or department.
`U_SEC_`	Universal	Security	**The "Bridge":** Consolidating roles from multiple domains.
`DL_SEC_`	Domain Local	Security	**The "What":** Assigning access to a specific resource.
`G_DIST_`	Global	Distribution	**Email Only:** Departmental email list (no permissions).
`ADM_`	Global	Security	**Admin Only:** Groups with high-level privileges (e.g., Server Admins).

Prefix Based Naming - Examples

Case A: Healthcare (The Clinic)

In a hospital setting, you might have different domains for "Clinical" and "Administrative" staff.

- **Global (Role):** `G_SEC_Nurses_DayShift` (Users: All day-shift nurses)
- **Universal (Consolidated):** `U_SEC_All_Medical_Staff` (Members: `G_SEC_Nurses` and `G_SEC_Doctors`)
- **Domain Local (Resource):** `DL_SEC_PatientPortal_Modify` (Permissions: "Modify" on the records server)

Case B: Education (The College)

- **Global:** `G_SEC_CS_Students_2026`
- **Global:** `G_SEC_Faculty_Adjunct`
- **Domain Local:** `DL_SEC_Lab04_Workstations_Login` (Grants the right to log into Lab 04)
- **Domain Local:** `DL_SEC_LectureNotes_Read` (Grants Read access to the faculty share)

Prefix Based Naming - Examples

3. The "Resource-First" Convention for DL Groups

For **Domain Local** groups, it is a best practice to name them after the **resource** first, then the **permission**. This keeps all related permissions sorted together in the list.

- `DL_SEC_SVR01_Accounting_R` (Read-only)
- `DL_SEC_SVR01_Accounting_M` (Modify/Write)
- `DL_SEC_SVR01_Accounting_F` (Full Control)

by using `_R`, `_M`, and `_F` as suffixes, an admin can quickly see if they accidentally put a user in both "Read" and "Modify" groups for the same folder.

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4. Naming for "Special" Group Types

Sometimes you need groups that don't fit the standard AGUDLP role.

- **Application Groups:** `APP_Confluence_Admins` or `APP_Salesforce_Users`.
- **Device Groups:** `DEV_Laptops_Win11` or `DEV_Printers_Color_Floor2`.
- **Service Accounts:** `SVC_SQLServer_RunAs` (Used to manage the "Log on as a service" right).

Group Naming Summary

Summary:

- **No Spaces:** Use underscores (_) or hyphens (-).
- **Unique Prefixes:** Always start with the scope/type (e.g., `G_SEC_`).
- **No Direct Users in Universal:** Only put Global groups into Universal groups.
- **Case Consistency:** Stick to either all CAPS or *PascalCase* (e.g., `G_Sec_Accounting`).
- **NetBIOS Awareness:** Keep the first 15 characters unique in case the environment still supports legacy systems.

Computer Accounts

- Windows computers require computer accounts in the domain before anyone can use them to logon to the domain
- A computer account is either created ***before*** a computer joins the domain or ***while*** a computer joins the domain.
  - If the computer account is created *first* the computer will be linked to that object if it shares that objects name (hostname = samAccountName)

Computer Accounts

- Every member computer in an AD DS domain maintains a computer account with a username (sAMAccountName) and password, just like a user account does.
  - The computer generates password to use and changes its password in the domain approximately every 30 days. This process is initiated by the Client computer and the *Netlogon* service.
  - Computer passwords are exempt from the domain password policy.
- Computer accounts are used in the same ways that user accounts are used for users (as principals for policy to be applied).
- Each computer account has a GUID and a Security Identification (SID) and additional attributes.
- Computer accounts (non-DCs) are members of the Domain Computers security group.

Default Computer Account Location

- When Active Directory is installed on the first DC in a domain, Active Directory creates two containers for computer objects:
  - **Computers** (folder object; container)
  - **Domain Controllers** (OU).
- A special Group Policy Object (GPO) is applied to the Domain Controllers OU (Default Domain Controllers GPO**)**
- The Computers folder is the default location for all new computer objects in AD.

Joining the Domain

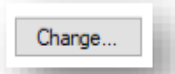
- Joining a computer must be
  - performed at the computer itself
  - done using WinRM and Powershell this can be done remotely
- The DNS for the computer must be pointed to the DNS that provides name resolution for the domain.
- To join a computer
  - Open advanced system properties (sysdm.cpl) → Computer Name tab → Change button → enter the appropriate domain you wish to join.
- A dialog box will prompt for an account with permissions to join the computer
  - You must have the credentials of a Domain Administrator (see below).
- Every Domain Users group member has automatically been given the right to join up to 10 workstations to the domain and create 10 corresponding computer accounts
  - This only applies to Client OSs not server versions.
  - Only administrators can create computer accounts for server systems
  - Can be disabled by policy/permissions (on the Computers AD container).

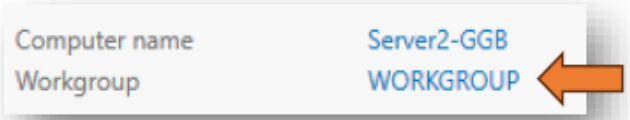
Resetting Computer Accounts

- You can reuse the same computer account for two different computers by resetting the computer account
- Resetting a computer account resets its password but maintains all its properties. With a reset password, the object is rendered available for use again. Any appropriately named computer can join the domain using that object.
- To reset a computer account, right-click the computer object and select Reset Account.

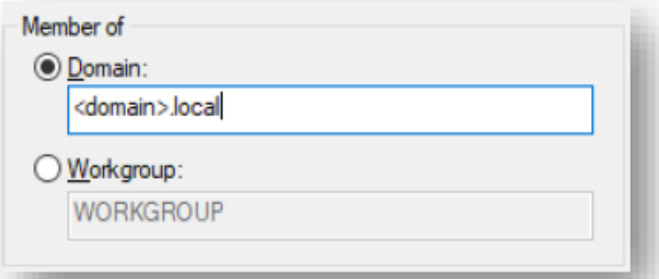
Do this…

- On Server2-XXX
  - Ensure that your preferred DNS is pointed to the IP of your DC (not 8.8.8.8)
  - Open advanced system properties (sysdm.cpl).
  - Computer Name → Change

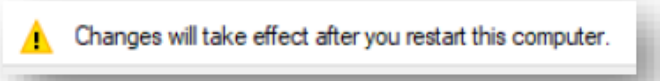




  - Member of → Domain → Your domain name.
  - Authenticate as administrator
    (you don't need the full legacy name as the domain was specified in the previous step)

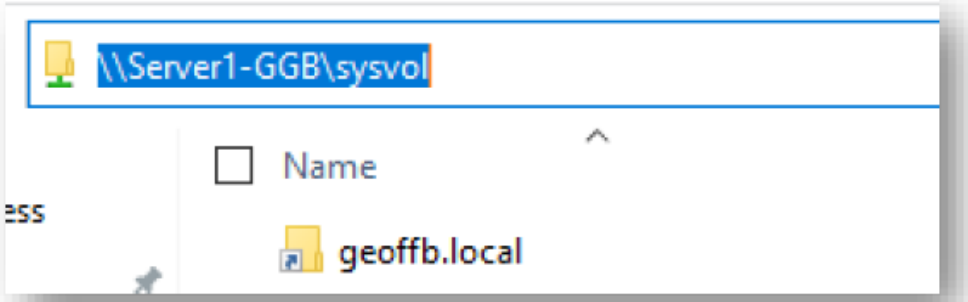


  - Restart



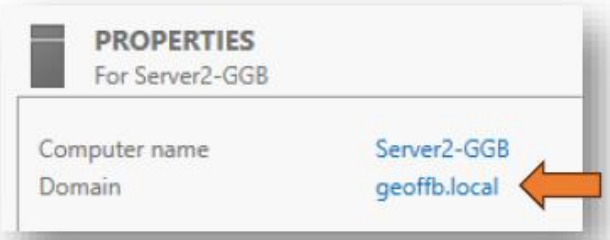
Examine changes post-domain joining

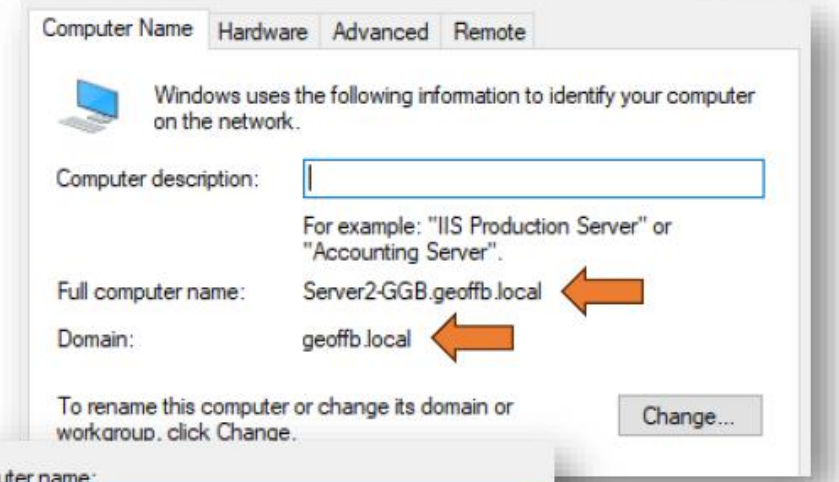
- There are a few things we want to look at now that Server2 is a domain member.
  - Its name (hostname) has changed
  - There are new objects in AD
  - Its local groups have changed
  - You can now log in as domain user/admins.
  - Group Policy is now applied from the domain (in addition to local policy) (you should be able to browse the \\DC\Sysvol share *without* having to re-authenticate)

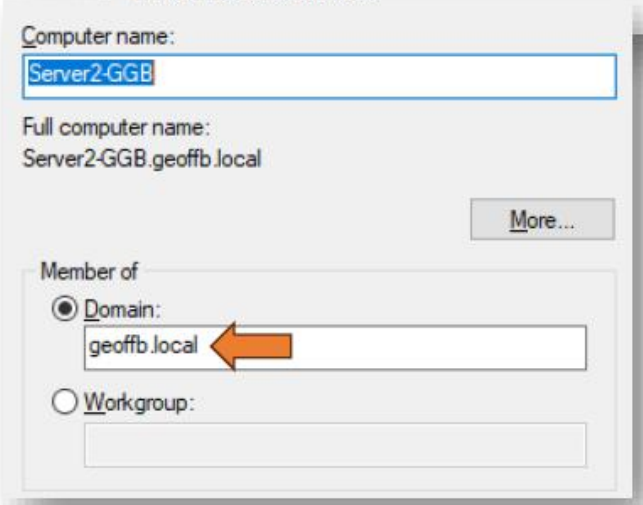


System Properties (changes)

- A few things to notice:
  - This computer is now a member server.
  - This is reflected is both server manager and advanced system properties.
  - `IPCONIG /ALL` reflects the change.








```
PS C:\> ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : Server2-GGB
   Primary Dns Suffix  . . . . . . . : geoffb.local
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No
   DNS Suffix Search List. . . . . . : geoffb.local
```

Default Computer Account Location

- When Active Directory is installed on the first DC in a domain, Active Directory creates two containers for computer objects:
  - Computers (folder object; container)
  - Domain Controllers (OU).
- A special Group Policy Object (GPO) is applied to the Domain Controllers OU (Default Domain Controllers GPO). There are no policies applied to member servers/clients specifically.
- The Computers *container* is the default location for all new computer objects in AD. It is a container (not an OU) and cannot have Group Policy applied to it directly.
  - To apply GPO to computers, you can use the Domain Container (inheritance) or move the computer objects to a new OU.

Do this...

- Powershell is the best way to see computer properties. Here's the command to do that.
- `Get-ADComputer -identity <samAccountName>`
- The samAccountName is the hostname with a trailing '$' on it.

Thing to note:

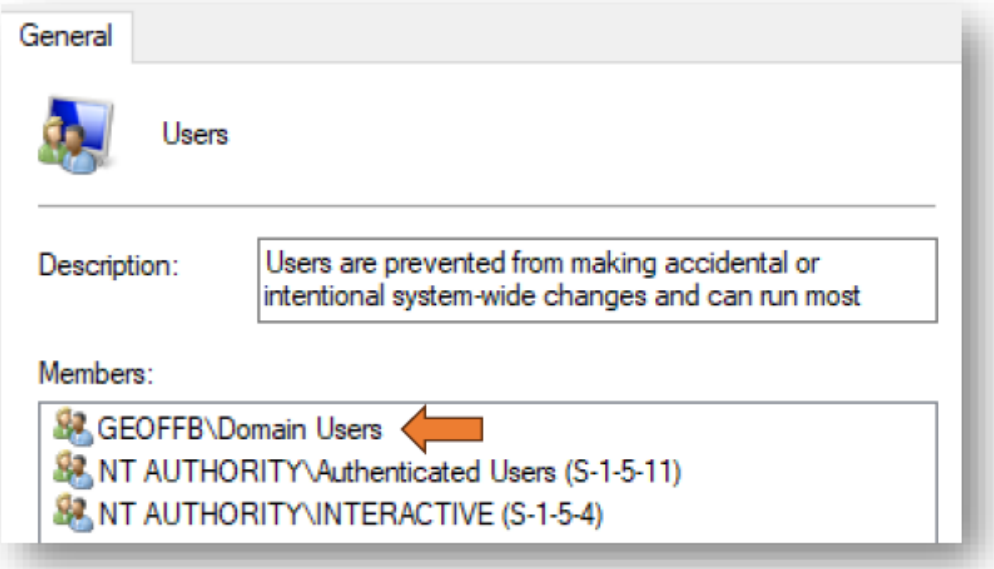
- Object DN
- FQDN
- Hostname
- GUID
- samAccountName
- SID

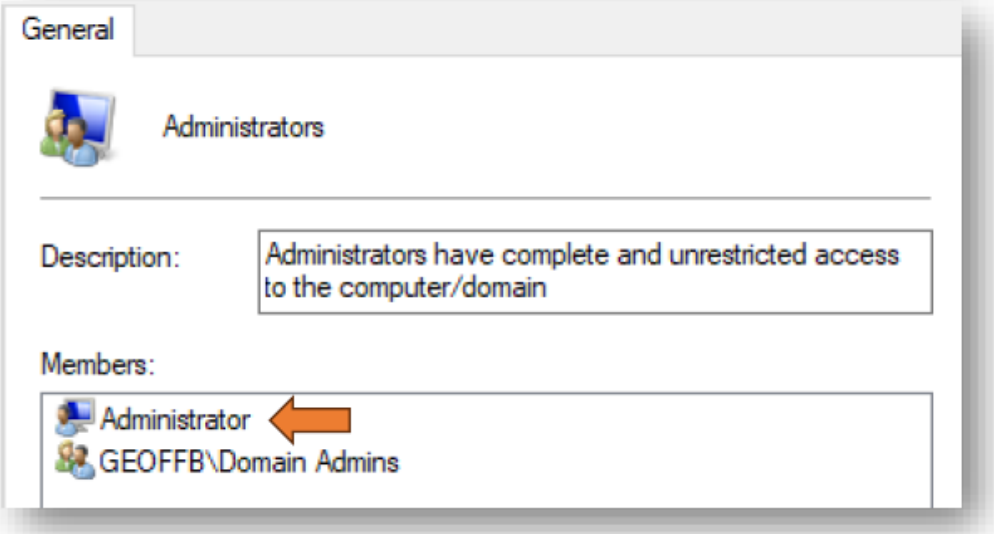
```
PS C:\> Get-ADComputer -identity server2-ggb$

DistinguishedName : CN=SERVER2-GGB,CN=Computers,DC=geoffb,DC=local
DNSHostName       : Server2-GGB.geoffb.local
Enabled           : True
Name              : SERVER2-GGB
ObjectClass       : computer
ObjectGUID        : 43ac9365-38b9-4875-b71b-dd2bdd6aa724
SamAccountName    : SERVER2-GGB$
SID               : S-1-5-21-3853701937-1181878592-2866956432-1125
UserPrincipalName :
```

Local Group Changes

- When a computer is Domain Joined, group nesting occurs.
- The global security group Domain Users is nested in the local Users group.
  - Any Domain User now has the rights of a standard User on this computer.
- The global security group Domain Admins is nested in the local Administrators group.
  - Any Domain Admin now has the rights of an Administrator on this computer.



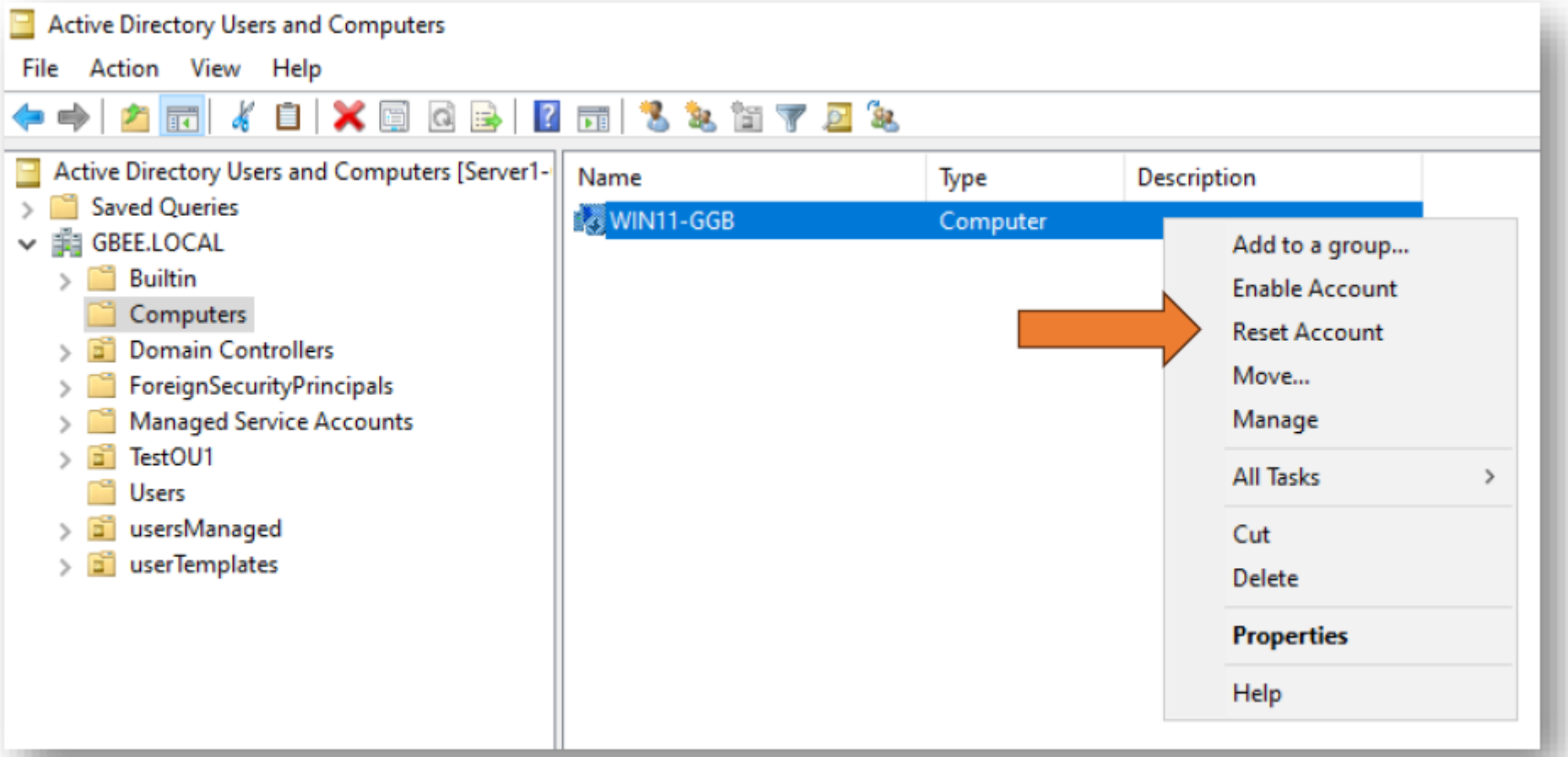


Resetting Computer Accounts

- You can reuse the same computer account for two different computers by resetting the computer account
- Resetting a computer account resets its password but maintains all its properties. With a reset password, the object is rendered available for use again. Any appropriately named computer can join the domain using that object.
- To reset a computer account, right-click the computer object and select Reset Account.

```
Reset-ComputerMachinePassword -Server “Server1-XXX" -Credential
<domain>\administrator
```

Resetting a Computer Account



Signs of Computer Account Problems

- Messages at logon that indicate a domain controller cannot be contacted, that the computer account might be missing, or that the trust (or secure channel) between the computer and the domain has been lost
- Error messages or entries in the Event Viewer that indicate similar problems or suggest that passwords, trusts, secure channels, or relationships with the domain or a domain controller have failed
- A computer account can not be located in Active Directory

Troubleshooting Computer Account Problems

- If the computer account exists in Active Directory, reset the account
- If the computer account is missing from Active Directory, create an account with the same name as the computer
- If the computer still belongs to the domain, disjoin it, by joining a workgroup
  - Then, Rejoin the computer to the domain

Do this…

To disjoin a computer from the domain, join it to a work group:

- Disjoin Server2 from the domain.
  - Advanced System Properties (sysdm.cpl) → Computer Name → Change
  - Select “workgroup” and enter a name → Ok → Restart.
- Look in AD and notice that the computer account is gone.
- To join a computer to the domain using PWSH.

```
Add-Computer `
      -ComputerName Server2-XXX `
      -LocalCredential Server2-XXX\Administrator `
      -DomainName <domain>.local `
      -Credential <domain>\administrator `
      -Restart
```

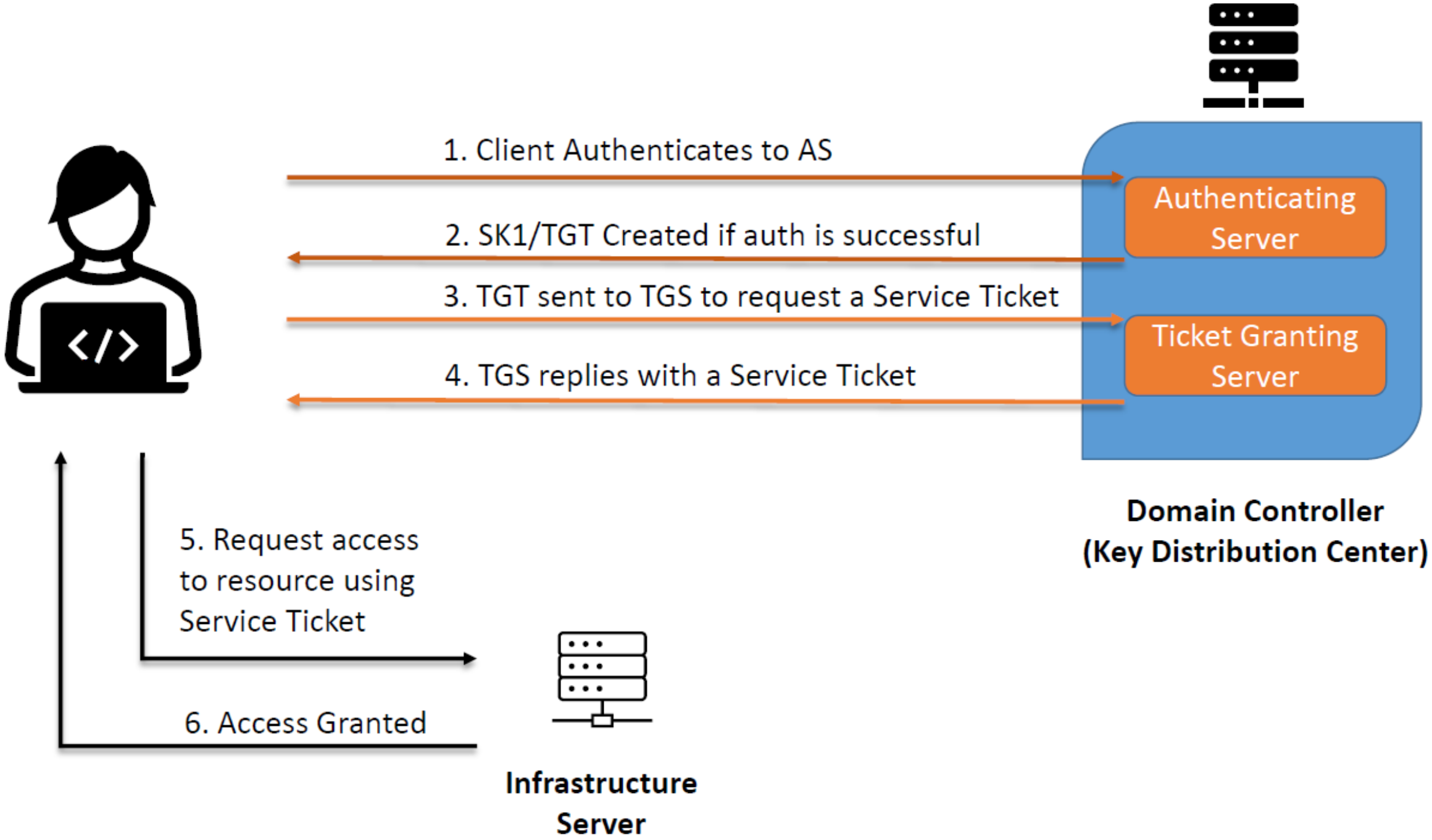
Kerberos

What is Kerberos?

- Kerberos is a computer network security protocol that authenticates service requests between two or more trusted hosts across a trusted or untrusted network.
- The default authorization technology used by Microsoft Windows
- two functions:
  - Authentication
  - Ticket-granting

How does it work (high-level)

1. Client Authenticates
   - Ask the Authenticating Server (AS) for a Ticket Granting Ticket (TGT).
2. The Key Distribution Center (KDC) checks to see if the Ticket Granting Server (TGS), and the client Information, are available.
   - If true, the AS will generate a session key (SK1, used for encryption)
   - The session key is used to create an encrypted TGT for the client.
3. Client decrypts the message, and extracts the SK1/TGT.
   - This gives the ability for the client to use the SK1/TGT during auth.
4. Client uses the TGT for access requests to the target server.
5. The TGS accepts the requests, checks the TGT for validity, and then provides a Service Ticket to be used for that session (to the requested resource)
   - This produces an SK2 which is used with the Service Ticket granted by the TGT.
6. Client uses the SK1 (TGT), to extract the SK2 (Service Ticket).
   - The Service Ticket is then sent to the target server.
7. Target server uses the SK2 (Service Ticket) to provide access to the requested resource.


1. Client Authenticates to AS
2. SK1/TGT Created if auth is successful
3. TGT sent to TGS to request a Service Ticket
4. TGS replies with a Service Ticket
Authenticating Server
Ticket Granting Server
Domain Controller (Key Distribution Center)
5. Request access to resource using Service Ticket
6. Access Granted
Infrastructure Server

Important Kerberos Settings

- The settings of Kerberos are controlled VIA Policy.
  Computer\Windows Settings\Security Settings\Account Policies\Kerberos Policy
- They are set on the Default Domain Policy and can ONLY be applied to the Domain container (not OUs,or sites).
- Notable settings
  - Maximum tolerance for computer clock synchronization
    - The maximum difference a computers clock can be off from that of the domain controller.
  - Maximum lifetime for user ticket
    - How long until your TGT expires
  - Maximum lifetime for service ticket
    - How long until your Service Ticket expires

Service Accounts

Service Accounts

- What is a service account?
  - The account under which a process or service runs.
  - Isolates privilege and rights for a Service.
  - May be Managed or Unmanaged

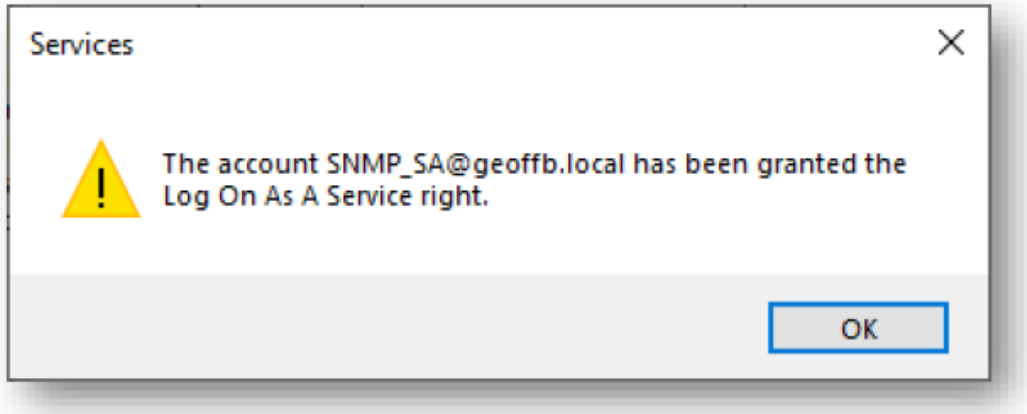
- Why use Service accounts?
  - You can create a **class** of domain accounts that can be used to manage and maintain services on local computers.
  - Unlike domain accounts in which administrators must manually reset passwords, the network passwords for these accounts are automatically reset(Managed Service Accounts Only).
  - Simplifies the management of Services
  - Administrative tasks for managed service accounts can be delegated to non-administrators (most services are running with too many privileges anyway).
- Service accounts are common in Server. Installing some applications will create their own service accounts.

What happens when I use a service account?

- That account is giving the “Log on as a Service” right VIA **local** group policy on the server in which the Service runs.
- Once configured, you must stop/start the service, so that it starts up using the new credential.
- The Service will run under the authority of the new Security Name Principal (SPN), inheriting the SPNs rights and permissions.

Do this...

- Create an OU called ServiceAccounts.
- Create a user called SNMP_SA in the ServiceAccounts OU
  - Give the account the log on name of “SNMPSA”
  - Give the account the description of “Unmanaged SNMP Service Account”.
- Open Services.msc and find the SNMP Trap Service.
  - Add SNMP_SA as the logon account for the SNMP Trap Service.
  - Set the password.
  - Restart the service.



Do this…

- Set the *‘Log on As’* back to Local Service
- Restart the SNMP Trap service

Delegation of OUs

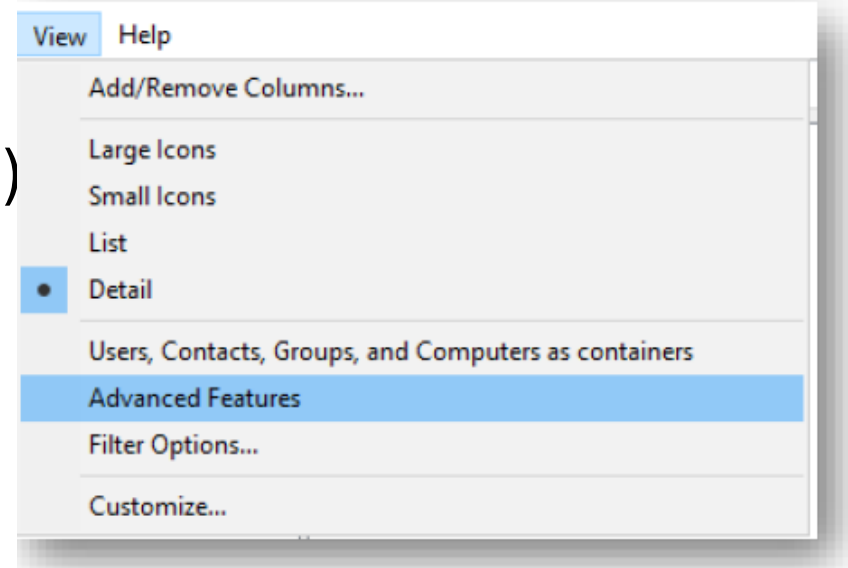
Delegation of Administration

(Delegate Control using OUs)

- You can use organizational units (OUs) to delegate the administration of objects (i.e. users or computers) within the OU to a designated individual or groups.
  - Occurs when a higher security authority assigns permissions to a lesser security authority
  - Allows you to distribute various roles and responsibilities of the IT staff
  - Can only be done at the OU/Container (or domain) level, not the object level
  - Due to the nature of parent-child inheritance for OUs, by delegating control of a parent OU, you are extending the delegation to all of the child OUs as well (similar to NTFS Permissions).
- Possible to delegate control over objects in the directory w/o changing the default control given to the service administrators

Delegation of Administration

(Delegate Control)



- Create a group structure (following best practices) access.
  - A DLG to hold the permissions
  - A GG to hold the user accounts
  - Nest the GG in the appropriate DLG.

- To delegate authority:
  - Right-click the OU and click Delegate control; Click Next after the wizard starts
  - Click the Add button and specify the accounts, groups, or computers to have the control
  - Select the tasks to delegate and click Next
- To verify permissions have been assigned, right-click the OU and access the Security tab (only viewable with the Advanced View turned on)

Do this…

- Create an OU called DelegationGroups.
- In the DelegationGroups OU create the following groups:
  - DLG_UsersManaged_AcntFull
  - DLG_UsersManaged_RstPwd
  - DLG_UsersManaged_PolMgr
  - GG_GeoffB_Deskside
  - GG_ GeoffB _ADObjectMgrs
- Nest the following groups:
  - GG_ GeoffB _Deskside → DLG_UsersManaged_RstPwd
  - GG_ GeoffB _ADObjectMgrs → DLG_UsersManaged_AcntFull
  - GG_ GeoffB _ADObjectMgrs → DLG_UsersManaged_PolMgr
- Delegate the Resetting of user passwords of users in the UsersManaged OU, to members of the GG_ GeoffB _Deskside group.
- Delegate the Create, Delete and Manage rights for users in the UsersManaged OU, to members of the GG_ GeoffB _ADObjectMgrs group.
- Delegate the Managing of Policy links applied to the UsersManaged OU, to members of the GG_ GeoffB _ADObjectMgrs group.